

Medical Cost Prediction in the United States

Project Report

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1 Introduction

With rising costs in healthcare, we want to develop systems to predict what factors contribute the most in the ever increasing medical charges. An article from the American Medical Association released in 2025 stated that in 2023, U.S. healthcare expenditures surged by 7.5% to reach a total of \$4.9 trillion. Noticeably many factors seem to drive the increase in medical charges. Cox et al. states that rising healthcare expenses are driven by a variety of factors which includes older populations, prevalence of chronic illnesses, costly new technologies, higher unit prices, and expanded insurance coverage. They also mention some of these trends occurring in the U.S. also reach other wealthy countries too.

Our goal is to identify and determine the degree of correlation between various predictors in the dataset. As such, we will construct a viable pipeline of models, from interpretable baselines (OLS) to flexible ensemble learners (Random Forest). We will use them to compare interpretability and utility, and learn the effect of various predictors. The dataset used is limited to 1,338 entries. Thus, it provides a relatively limited population of beneficiaries. Additionally, many factors are omitted, as there are only seven total predictors. Furthermore, this dataset does not provide temporal trends.

With this project, we aim to introduce models that effectively predict individual medical insurance charges from demographics/health attributes. Based on the results we obtain, we also aim to show that smoking and obesity together compound costs far beyond what either factor does alone. These results will be interpreted using a pipeline of various statistical models, with varying levels of assumptions and interpretability.

2 Data Description

We are utilizing a dataset consisting of 1338 observations of patient insurance costs based on seven variables. However, after identifying and removing a duplicate row, the actual count stands at 1337 unique observations. We will take each individual policyholder as an observational unit. As stated previously, this dataset acts as a snapshot of individual policies, distributed evenly across the United States.

The original dataset has non-numerical features where the features sex has values “male” and “female”, smoker has values “yes” and “no”, and region has values “southwest”, “northwest”, “southeast”, and “northeast”. Given this, we converted them such that all the features in the dataset now have numerical values as shown in Table 1.

Table 1. Variable Descriptions

Variable	Description	Type	Unit
<i>Charges</i>	Cost of policyholders medical insurance	Continuous $\in [1121.87, 63770.43]$	\$ USD
<i>Age</i>	Age of beneficiary	Continuous $\in [18, 64]$	Year
<i>BMI</i>	Body Mass Index of beneficiary	Continuous $\in [15.96, 53.13]$	$\frac{kg}{m^2}$
<i>Sex</i>	Sex of beneficiary	Binary $\in \{0, 1\}$	female=0, male=1
<i>Smoker</i>	Smoking status of beneficiary	Binary $\in \{0, 1\}$	non-smoker=0, smoker=1
<i>Region</i>	Region of residence within the US	Categorical $\in \{0, 1, 2, 3\}$	Northeast=0, Northwest=1, Southeast=2, Southwest=3
<i>Children</i>	Beneficiaries number of children/dependents	Categorical $\in \{0, 1, 2, 3, 4, 5\}$	Children

3 Exploratory Data Analysis (EDA)

The age feature is roughly uniformly distributed with values between 18 and 64 years old. Age is approximately symmetric with a noticeable yet negligible spike at age 18-19 and is a platykurtic distribution (i.e. less kurtosis) with no strong central tendency. BMI is approximately normally distributed with the mean centered around 30.66, slightly above the clinically defined threshold for adult obesity of $BMI \geq 30$ according to the Centers for Disease Control and Prevention (CDC) and World Health Organization (WHO). The charges label is significantly right skewed with a median price of \$9,382 but a mean price of \$13,270 pulled upward by a high-cost tail. This bimodal distribution shown in the histogram and pair plot in **Figure A1** is the first hint that two fundamentally different cost populations exist in the data.

We decided to run a Spearman correlation over the commonly used Pearson correlation. Pearson is used for linear, continuous, and normally distributed data without outliers. However, Spearman is used for monotonic relationships (variables move together but not linearly), ordinal data, skewed data, or when outliers are present. Spearman is non-parametric (ranking-based), making it more robust (Mukaka, 2012). We noticed in **Figure A2**, smoking is by far the strongest predictor of charges ($\text{corr} = 0.66$) This is shown in group level statistics for smokers vs non-smokers where the average charge for smokers (mean of \$32,050; median of \$34,456) is around four times the average charge for non-smokers (mean of \$8,434; median of \$7,345). We also noticed that smokers heavily amplified the charges when factoring in obese and non-obese patients as **Figure A4** shows that non-smoker obese and non-smoker non-obese groups have similar and relatively low charges (medians around \$7,000 - \$8,000). Smoker non-obese individuals jump to a median of around \$20,500, and smoker obese individuals reach an even higher median of around \$40,500. This interaction effect suggests that smoking and obesity together compound costs far beyond what either factor does alone, which is a critical finding for any predictive model. In the Spearman correlation, the second most correlated variable, age, clearly influences charges, with median costs rising steadily from \$9,398 for the 18–30 group to \$21,063 for those 60 and older. When visualizing the age and charges stratified scatter plot, there appear to be three distinct cost bands across all ages with a low band (non-smokers), a high band (smokers), and a middle band (combination of both smokers and non-smokers). These bands are shown in **Figure A3**.

The data shows that for predictors, smoking status is the dominant driver of medical insurance cost, which is responsible for a near four times premium in charges and creating a bimodal cost distribution. Age with charges roughly doubling across the age spectrum. BMI matters mainly in interaction with smoking rather than in isolation. Sex, region, and children are statistically significant but practically weak predictors.

4 Modeling Methodology

This section presents the four regression models estimated in the analysis, progressing from a simple baseline to a high-capacity ensemble method. Each model is motivated by specific statistical or theoretical considerations, and together they form a coherent pipeline that systematically improves predictive accuracy.

4.1 Baseline Ordinary Least Squares (OLS)

The baseline OLS directly takes the original 6 features without any transformations, derived variables, or interaction terms and is explicitly stated in (1):

$$\widehat{\text{charges}} = \beta_0 + \beta_1 \cdot \text{age} + \beta_2 \cdot \text{sex} + \beta_3 \cdot \text{bmi} + \beta_4 \cdot \text{children} + \beta_5 \cdot \text{smoker} + \beta_6 \cdot \text{region} + \varepsilon \quad (1)$$

Charges is the dependent variable (annual medical insurance charges in USD), β_0 is the intercept, $\beta_1 - \beta_6$ are slope coefficients associated with each predictor, and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is the random error term. OLS regression tends to rely on classical Gauss-Markov conditions to guarantee that coefficient estimates are best linear unbiased estimators (Frost, 2018). In this model, many key assumptions seem to be violated such as linearity where given the right-skewed, bimodal distribution of charges driven largely by the smoker subgroup, the linearity assumption is expected to be violated thus producing poor predictions at the upper tail. Another assumption violated is homoscedasticity where variance of the random error term conditioned on the predictors should be constant across all observations. However, insurance charges are known to be heteroscedastic where variance scales with charge level, implying larger residuals for high-charge smoker groups. The assumption of independence is satisfied as observations are drawn independently.

The model parameters are estimated using Ordinary Least Squares via statsmodels (sm.OLS). OLS minimizes the residual sum of squares in closed form via the normal equations in regression. Standard errors are computed under the non-robust homoscedastic assumption, consistent with a diagnostic baseline. While heteroscedasticity-robust standard errors were not employed here so that model limitations are visible. The baseline model is intentionally simple and is expected to exhibit deficiencies such as underfitting on high-charge cases since extreme charge values have excessive leverage on the regression hyperplane thus inflating RMSE. There were also no nonlinearity or interaction effects. The positive skew of charges also propagated into the residuals which violates OLS assumptions inflating prediction intervals.

4.2 Log-Linear OLS

Since insurance charges are strictly positive and right-skewed, taking the natural logarithm of charges compresses the heavy right tail, producing a distribution closer to normality. It also creates proportional (as opposed to additive) error variance, thus reducing

heteroscedasticity. Additionally, this new OLS model introduces feature sets designed to capture known nonlinearities and heterogeneous treatment effects.

age^2 is a quadratic age term that allows the marginal cost of aging to accelerate over time. *obese* is a binary indicator for $BMI \geq 30$ made to capture a discrete threshold effect independent of continuous BMI. Finally, the smoker interaction terms (*smoker_bmi*, *smoker_age*, *smoker_obese*) were added as the health risk of obesity and elevated BMI is substantially amplified by smoking. Interaction terms explicitly model the differential effect of BMI and age on charges conditional on smoking status.

This log-linear model (2) is:

$$\begin{aligned} \ln(\widehat{\text{charges}}) = & \beta_0 + \beta_1 \cdot \text{age} + \beta_2 \cdot \text{age}^2 + \beta_3 \cdot \text{sex} + \beta_4 \cdot \text{bmi} + \beta_5 \cdot \text{children} \\ & + \beta_6 \cdot \text{smoker} + \beta_7 \cdot \text{smoker_bmi} + \beta_8 \cdot \text{smoker_age} + \beta_9 \cdot \text{smoker_obese} \\ & + \beta_{10} \cdot \text{obese} + \beta_{11} \cdot \text{region} + \varepsilon \quad (2) \end{aligned}$$

Notice that coefficient interpretation follows the semi-elasticity framework where for a continuous predictor X_i . The percentage change in charges associated with a one-unit increase in X_i , holding all else constant, is approximately $(e^{\beta_i} - 1) \times 100\%$. The intercept e^{β_0} represents the geometric mean baseline charge when all predictors are zero. Key assumptions is this model assumes $E[\ln(\text{charges}) \mid X]$ is linear in parameters. This model has a much better, yet not perfect, approximate normality of log-residuals. The proportional error structure of the log model is better suited to insurance data than constant additive variance.

Note that when predicting charges on the dollar scale, fitted values are obtained by exponentiating logscale predictions: $\hat{y}_i = e^{\ln(\widehat{\text{charges}})_i}$. However, due to Jensen's inequality, this naive back-transformation yields a median predictor rather than the conditional mean. A Duan smearing factor or lognormal bias correction $e^{(\sigma^2/2)}$ can be applied to recover the conditional mean (Duan, 1983). In this analysis, naive back-transformation is used throughout for consistency across models, with the understanding that reported metrics reflect median-targeting performance. By modeling the log-transformed outcome with the enriched feature set, Model (2) is expected to outperform the baseline on every metric including dramatically lower RMSE and MAE due to the removal of extreme residuals from the heavy-tailed raw charge distribution. Additionally higher R^2 values imply additional explanatory power of interaction terms and nonlinear aging effects.

4.3 Linear Mixed-Effects Model (LME)

In Model (2), region was included as a fixed effects term which treats geographic variation in charges as a single additive constant. However, treating region as a fixed effect has a limitation, where regional intercepts may be better characterized as draws from a common distribution rather than being exactly identified from the data. The Linear Mixed-Effects (LME) model handles this by partitioning the intercept into a population-level fixed effect and a region specific random deviation which borrows statistical strength across regions through partial pooling or shrinkage estimation (Vasishth, 2022).

For $j \in \{0, 1, 2, 3\}$ indexing the four geographic regions. The LME model (3) is:

$$\begin{aligned} \ln(\widehat{\text{charges}}) &= (\beta_0 + u_j) + \beta_1 \cdot \text{age} + \beta_2 \cdot \text{age}^2 + \beta_3 \cdot \text{sex} + \beta_4 \cdot \text{bmi} + \beta_5 \cdot \text{children} + \beta_6 \cdot \text{smoker} \\ &\quad + \beta_7 \cdot \text{smoker_bmi} + \beta_8 \cdot \text{smoker_age} + \beta_9 \cdot \text{smoker_obese} + \beta_{10} \cdot \text{obese} + \varepsilon_j \\ u_j &\sim \mathcal{N}(0, \sigma_u^2) \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_\varepsilon^2) \quad \text{Cov}(u_j, \varepsilon_j) = 0 \quad j \in \{0, 1, 2, 3\} \text{ (regions)} \end{aligned} \quad (3)$$

Here, β_0 is the grand-mean intercept (fixed), u_j is the region-specific random intercept capturing unobserved geographic heterogeneity, σ_u^2 is the between-region variance, and σ_ε^2 is the within-region residual variance. The main assumptions are normality of random effects and of residuals along with balanced group sizes which ensures that shrinkage does not heavily distort any single region's estimate. There also needs to be independence of random effects and predictors which implies unobserved regional factors are uncorrelated with the included predictors.

The model is fitted using Restricted Maximum Likelihood (REML) via `statsmodels.MixedLM`. REML is preferred over the full maximum likelihood for variance component estimation yielding less-biased estimates of σ_u^2 and σ_ε^2 . Fixed-effect coefficient estimates and their standard errors are obtained as a byproduct of the REML optimization. The intraclass correlation coefficient (ICC) is computed as $\text{ICC} = \sigma_u^2 / (\sigma_u^2 + \sigma_\varepsilon^2)$, quantifying the proportion of total log-charge variance attributable to regional clustering (Liljequist, 2019). Model (3) and (2) share the same fixed-effect specification (excluding region as a predictor in LME, where it enters as a grouping variable).

4.4 Random Forest (Ensemble Learning)

While the OLS and LME models are constrained to linear combinations of the input features, the true relationship between covariates and insurance charges may exhibit complex nonlinearities and high-order interactions that cannot be captured even with manually engineered polynomial and interaction terms. Random Forest, a bagged ensemble of decision trees, is introduced as a nonparametric alternative that provides a robust, automated framework for modeling complex, nonlinear relationships and interpreting variable importance, while establishing a high-performance benchmark for model validation.

Random Forest constructs B independent decision trees with each trained on a bootstrap resample of the training data (bagging). In regression cases the final prediction is the average of all tree predictions written as (4).

$$\hat{F}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x) \quad (4)$$

Here $\hat{f}_b(x)$ is the prediction of the b -th tree. The log-transformed target $\ln(\text{charges})$ is used as the response variable, matching the formulation of Models (2) and (3), and enabling fair metric comparisons after back-transformation. Many strict assumptions from previous models are relaxed in Random Forests where the main ones are a sufficient sample size and no explicit error distribution such that it doesn't assume normality, homoscedasticity, or linearity, making it robust to the distributional violations.

The Random Forest model was implemented using `sklearn.ensemble.RandomForestRegressor`

with 44 estimators, derived from the models loss curve yielding minimum MSE value (see **Figure A5**). A test split of 20% was used. Five-fold cross-validation was applied to the training set using `cross_val_score` with R^2 scoring, yielding an unbiased estimate of out-of-sample performance robust to the specific train/test partition.

5 Evaluation Framework

We evaluate our four-model pipeline through a comparison of R^2 , RMSE, MAE, and MAPE. Supplemental information in the form of Actual vs. Predicted, Residuals vs. Fitted Values, and Q-Q Plots are included in the appendix. For the Random Forest analysis, a training/testing is split with the standard 80%/20% framework. It is further evaluated with a confusion matrix to test for bucket accuracy.

6 Results

Table 2. Model Performance Comparison

Model	R^2	RMSE (\$)	MAE (\$)	MAPE (%)
Baseline OLS	0.7506	6,046	4,175	42.24
Log-Linear OLS	0.8333	4,942	2,482	17.51
Linear Mixed-Effects	0.8336	4,938	2,468	17.39
Random Forests	0.8971	4,348	2,060	20.41

6.1 Baseline OLS Results

The baseline OLS model achieves an R^2 of 0.7506, an RMSE of \$6,046, an MAE of \$4,175, and a MAPE of 42.24%. While the R^2 value indicates that roughly 75.1% of the variance in raw charges is explained by the model, the large error magnitudes and high MAPE signal considerable predictive inaccuracy, particularly in the upper tail of the charge distribution. The global F -statistic of 667.0 ($p < 0.001$) shows that the model is jointly significant.

Table B1 presents the estimated coefficients, standard errors, t -statistics, and 95% confidence intervals. Smoking status is by far the dominant predictor, with a coefficient of \$23,820 ($p < 0.001$). This indicates that (conditional on all other covariates) smokers has approximately \$23,820 more in annual charges than non-smokers. Age contributes a statistically significant positive effect of \$257.20 per additional year ($p < 0.001$), which is consistent with increasing healthcare over the life cycle. BMI also exerts a significant positive effect of \$332.60 per unit increase ($p < 0.001$). Each additional dependent child is associated with \$478.77 in higher charges ($p < 0.001$). In contrast, sex ($\beta = -129.40$, $p = 0.698$) is not statistically significant, and region carries a small negative coefficient ($\beta = -354.01$, $p = 0.020$) that is significant but substantively modest.

Despite its statistical significance of many features, the model has clear diagnostic deficiencies which are consistent with the violated Gauss-Markov assumptions mentioned in Section 4.1. The Omnibus test statistic of 298.47 ($p < 0.001$) and a Jarque-Bera statistic of 711.71 ($p < 0.001$) indicate strong non-normality of residuals, with residual skewness

of 1.206 and excessively high kurtosis of 5.637. This is driven primarily by the heavy right tail of the raw charges distribution among smokers where the extreme charge values have disproportionate leverage on the regression hyperplane. These diagnostic findings motivate the log-transformation and enriched feature set employed in Model (2).

6.2 Log-Linear OLS Results

Applying the natural logarithm to the outcome variable and augmenting the feature set with an age-squared term, an obesity indicator, and three smoking interaction terms (*smoker_bmi*, *smoker_age*, *smoker_obese*) provides substantial improvements across every metric. Model (2) achieves an R^2 of 0.8333, an RMSE of \$4,942, an MAE of \$2,482, and a MAPE of 17.51%. The RMSE, MAE, and MAPE decline by approximately 18%, 41%, 59% respectively relative to the baseline reflecting the improved predictive accuracy that follows from modeling proportional rather than additive error structure.

The coefficient estimates and their semi-elasticity interpretations are summarized in **Table B2**. The grand-mean intercept $e^{6.8323} \approx \$927.32$ represents the geometric mean baseline charge when all predictors are zero. Each additional year of age raises charges by approximately 6.18% ($\beta_{\text{age}} = 0.0599$, $p < 0.001$), though the statistically significant negative quadratic term ($p < 0.001$) indicates that this marginal cost decelerates at higher ages. The sex coefficient implies that female beneficiaries incur charges approximately 8.15% ($p < 0.001$) lower than male beneficiaries. The continuous BMI coefficient ($p = 0.872$) and obesity binary indicator ($p = 0.947$) are both statistically insignificant in isolation, indicating that the marginal effect of BMI on charges operates primarily through its interaction with smoking rather than as a direct additive effect.

Each additional child is associated with a 10.2% increase in charges ($p < 0.001$). The smoker main effect implies that, for a non-obese smoker with average BMI and age, charges are approximately 644.92% higher than an otherwise identical non-smoker ($p < 0.001$). The interaction *smoker_bmi* ($p = 0.008$) reveals that each additional BMI unit amplifies the smoking premium by a further 1.86%, capturing the compounding health cost of high body weight among smokers. The *smoker_age* interaction ($p < 0.001$) implies a 3.29% reduction in the smoker premium per additional year of age. This reflects that the differential between smoker and non-smoker charges narrows somewhat as both populations accumulate age-related costs. The *smoker_obese* interaction ($p < 0.001$) indicates that smokers who are obese incur charges approximately 67.1% higher than smokers of equivalent BMI who are not classified as obese, identifying a particularly high-risk subgroup. Finally, the region coefficient ($p < 0.001$) implies a 5.18% decline in charges per region index unit, capturing modest geographic variation.

However, note that the diagnostic statistics indicate residual non-normality. Residual skewness rises to 3.23 from 1.21 under the baseline, and the Jarque-Bera statistic of 10,437 ($p < 0.001$) strongly rejects normality of log-scale residuals. The condition number of 4.1×10^4 also flags potential multicollinearity among the interaction terms. This is expected given that *smoker_bmi*, *smoker_age*, and *smoker_obese* are derived from the same binary smoker variable.

6.3 Linear Mixed-Effects Model Results

Replacing the fixed region term from Model (2) with a random intercept produces a negligible but consistent improvement across all metrics as shown in **Table B3**. Model (3) achieves an R^2 of 0.8336, an RMSE of \$4,938, an MAE of \$2,468, and a MAPE of 17.39%. The gains over the log-linear OLS are very minor as the fixed-effect coefficient estimates are nearly unchanged given that both models share the same predictor structure, and the primary difference lies in how geographic variation is partitioned rather than in additional explanatory power.

The variance components estimated via REML are $\hat{\sigma}_u^2 = 0.0045$ and $\hat{\sigma}_\varepsilon^2 = 0.1434$, yielding an ICC = 0.0306. This indicates that only 3.06% of the total variance in log-charges is attributable to between-region clustering. This confirms that geographic region is a statistically significant yet minor source of heterogeneity in this dataset. The estimated region-level random intercepts are $\hat{u}_{NE} = 0.0778$, $\hat{u}_{NW} = 0.0270$, $\hat{u}_{SE} = -0.0446$, and $\hat{u}_{SW} = -0.0602$. These intercepts capture unobserved geographic heterogeneity after conditioning on all fixed-effect predictors. Notice that the northern regions carries the positive deviation, while the southern regions carries the negative deviation, implying moderately higher baseline charges in northern states relative to southern ones.

The fixed-effect estimates and significance levels in Model (3) are very similar to Model (2), which validates that the log-linear specification captures the dominant structure of the data. The LME framework remains the more theoretically appropriate specification when region is conceptualized as a sample of a broader population of geographic units rather than as a set of exactly-identified levels.

6.4 Random Forest Results

The Random Forest ensemble trained on the log-transformed target and evaluated on a held-out 20% test partition achieves an out-of-sample R^2 of 0.8971, an RMSE of \$4,348, an MAE of \$2,060, and a MAPE of 20.41%. The R^2 , RMSE, and MAE each represent improvements over all preceding models, consistent with the ensemble’s capacity to capture nonlinear structure and high-order interactions beyond what the manually engineered polynomial and interaction terms in Models (2) and (3) can fully represent. The MAPE of 20.41% is, however, marginally higher than those of Models (2) and (3) (17.51% and 17.39% respectively), reflecting the Random Forest’s relative weakness in regions of sparse neighbor density at extreme charge values. The Random Forest nonetheless remains the strongest overall model, as large absolute dollar errors are minimized across the distribution. This is shown in **Table B4**

The number of estimators was set to 44 such that the training and testing R^2 scores were improved **Figure A5**. Five-fold cross-validation applied to the training partition yields a mean R^2 of 0.781 (std = 0.026), indicating stable generalization across folds. Cross-validation was performed on `X_train` exclusively, ensuring no leakage of test data into the validation process. The feature importance rankings confirm that *smoker_bmi* is the single most informative predictor at 28.4%, followed by *age* (20.8%) and *age*² (17.2%), which together account for 66.4% of the total importance mass. This corroborates the log-linear OLS finding that the interaction between smoking status and body weight, alongside the nonlinear age trajectory, drives the majority of variance in insur-

ance charges. Full feature importances, SHAP value decompositions, and the cost-bucket confusion matrices are provided in the appendix (**Figures A6, A7**).

Figure 1 shows the actual data points against the predicted regression curves from the log-linear OLS (dashed) and the Random Forest (solid), plotted against the smoker–BMI interaction. For non-smokers, both regressions yield a broadly flat charge profile across the BMI range, consistent with the finding that BMI is not a meaningful independent predictor of charges in the absence of smoking. For smokers, a notable discontinuity is visible in the OLS regression at BMI = 30, the obesity threshold. This is a modelling artefact arising from the inclusion of the binary *obese* and *smoker_obese* indicators: because OLS assigns fixed coefficients to these terms, predicted charges jump discretely at the threshold rather than transitioning smoothly. Replacing the binary obesity term with the continuous BMI term would eliminate this discontinuity but at the cost of predictive accuracy, interpretability, and comparability with the Random Forest specification. The Random Forest regression instead reveals a broadly continuous increase in predicted charges for smokers across BMI $\in (15, 30)$, followed by a further continuous increase above the obesity threshold, with a detectable step at BMI = 30 that is nonetheless smoother than the OLS counterpart. This behaviour characterises the obesity effect as a hybrid of continuous and threshold dynamics: smokers exhibit a mild positive trend in charges as BMI increases, but those surpassing the obesity threshold incur substantially elevated charges relative to those below it.

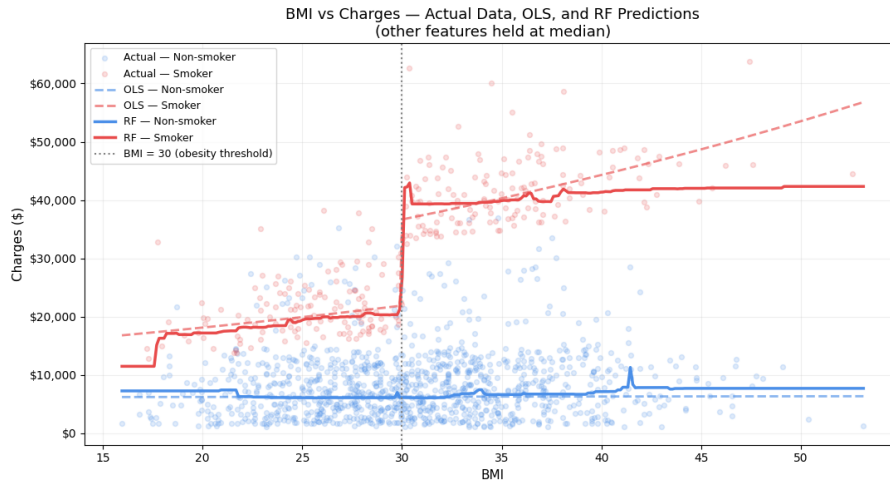


Fig. 1. Scatterplot of actual values, OLS Predicted Regression, RF Predicted Regression

6.5 Diagnostics

These diagnostics are based off of the plots presented in **Figure A8** and **Figure A9**

Model 1 - Baseline OLS The actual vs predicted plot reveals a clear fan-shaped dispersion which indicates heteroscedasticity that worsens at higher charge values. The residuals vs. fitted plot confirms this, with a pronounced LOWESS curve and residuals inflating beyond \$20,000 fitted values. The Q-Q plot ($r = 0.9482$) shows moderate normality in the center but heavy tails, reflecting skewed errors. The performance is clearly the weakest across all metrics ($R^2 = 0.7506$, RMSE = \$6,046, MAPE = 42.24%).

Model 2 - Log-Linear OLS Applying a log transformation substantially compresses the spread in the actual vs. predicted plot. Residuals are more homoscedastic at low fitted values but show a slight negative curvature in the LOWESS trend which suggests mild non-linearity remains. The Q-Q plot ($r = 0.7769$) indicates right-skewed residuals with a notable upper-tail departure. Performance improves dramatically ($R^2 = 0.8333$, RMSE = \$4,942, MAPE = 17.51%).

Model 3 - Linear Mixed-Effects Diagnostics are nearly identical to Model 2 which reflects comparable model structure. The actual vs. predicted scatter and residual plots exhibit a similar downward LOWESS curve, and the Q-Q plot ($r = 0.7734$) mirrors the log-linear OLS departure pattern. Metrics are essentially equivalent ($R^2 = 0.8336$, RMSE = \$4,938, MAPE = 17.39%). This suggests the mixed-effects component offers negligible improvement here.

Model 4 - Random Forest This model yields the tightest actual vs. predicted alignment, with observations clustering closely around the identity line across the full charge range. The residuals vs. fitted plot is the most homoscedastic of all four models, with the LOWESS curve lying nearly flat at zero. However, the Q-Q plot ($r = 0.7983$) still shows moderate tail departures, suggesting some non-normality in residuals. It does achieve the best overall performance ($R^2 = 0.8971$, RMSE = \$4,348, MAE = \$2,060, MAPE = 20.41%), though its higher MAPE relative to the log-linear models may reflect sensitivity to extreme values.

7 Discussion and Conclusion

Our four-model pipeline (OLS $\rightarrow$ Log-Linear OLS $\rightarrow$ LME $\rightarrow$ RF) demonstrates a clear interpretability-flexibility tradeoff where as structural assumptions relax and predictive accuracy improves, thus interpretability reduces. The baseline OLS was the most transparent but weakest performer, while RF delivered the strongest results at the cost of requiring additional diagnostic tools, such as SHAP values, to communicate its effects. Despite RF’s tendency to underestimate extreme values, our confusion matrix confirms high classification accuracy across all cost buckets, none falling below 85%. Across all models, smoking status emerges as the dominant cost driver, particularly through its interaction with BMI which is an effect that grows cleaner and more pronounced as the pipeline progresses. Age contributes a consistent, moderate positive effect, while region, sex, and number of children are statistically present but practically negligible.

Several limitations constrain these findings. The dataset’s 1,337 observations and only seven predictors reduces critical clinical details that would’ve been crucial in analysis, such as pre-existing conditions, procedure-level data, and causal/temporal trends, thus limiting the model’s real-world generalization. Additionally, ordinal encoding of the *region* feature imposes a false ranking. Future work should replace this with one-hot encoding. Also, a two-stage modeling approach could better handle high-cost outliers, and enriching the dataset with additional features and observations would strengthen predictive validity across the full charge distribution. Finally we should apply a consistent train/test split across all models to enable fair out-of-sample comparison.

Contributions

Each individual worked mostly on the items as labeled below. However, this does not include editing and other minor work done, as there was some sharing between all topics.

Dave Melkani primarily worked on the EDA, as well as constructing the OLS, Log-linear, and LME models. Eric Ness worked mostly on the Random Forest model and its interactions with the OLS. Analyses of the results were conducted jointly. Similarly, the presentation and the report were written jointly.

LLM usage was imperative in saving time and learning about model construction. For coding, LLM assisted with refining and correcting code. It additionally was able to improve the visual aspects of the plots used. LLMs also provided assistance with converting information between formats (e.g. creating LaTeX code for pre-written report materials, expediting citations, etc). Lastly, LLMs helped with brainstorming ideas and learning about some of the techniques used in this project.

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A Appendix (Plots)

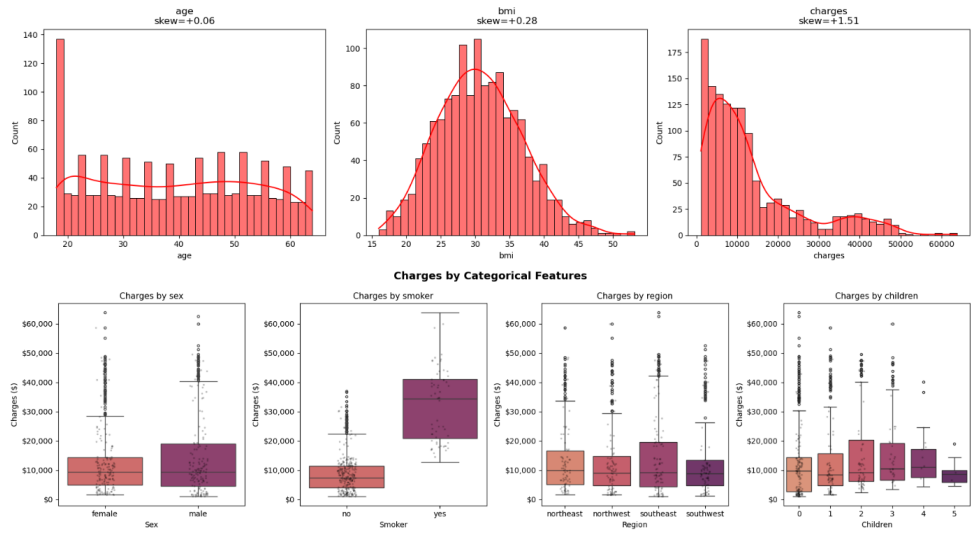


Fig. 1. Numeric Feature Distributions with KDE & Normal Fit and Charges by Categorical Features

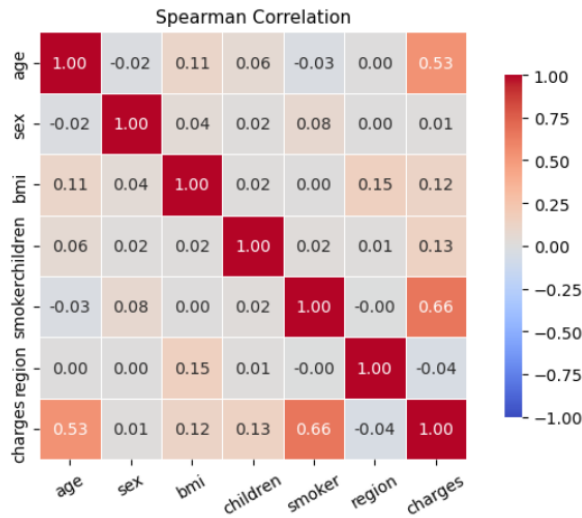


Fig. 2. Spearman Correlation Analysis

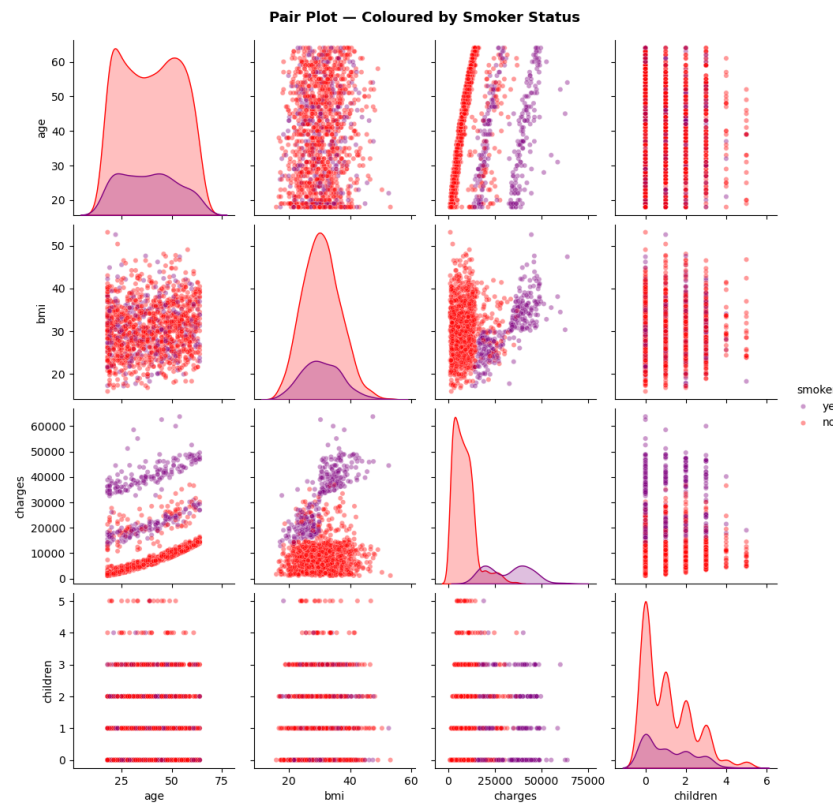


Fig. 3. Pair Plot Colored By Smoker Status

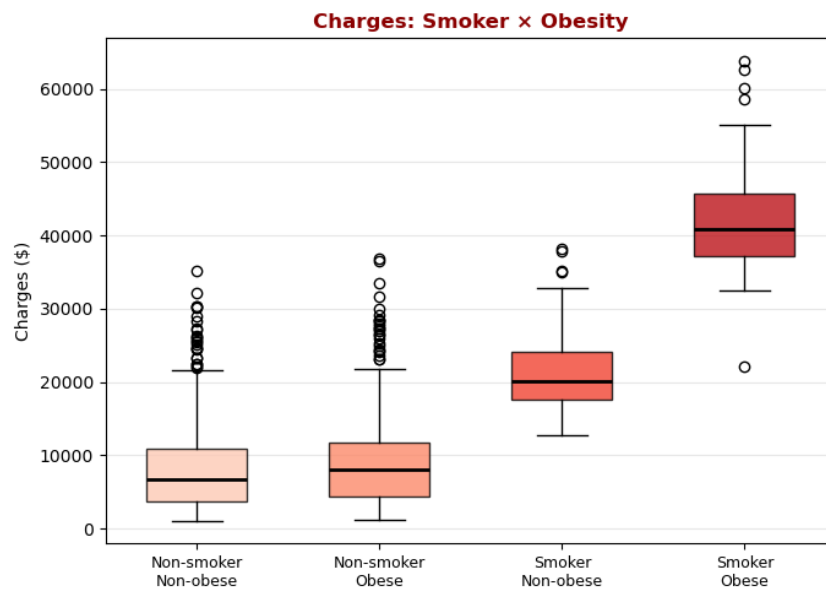


Fig. 4. Charges by Smoker × Obesity

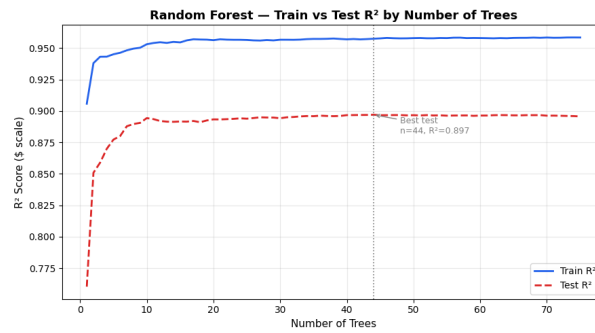


Fig. 5. Training and Testing R^2 Scores.

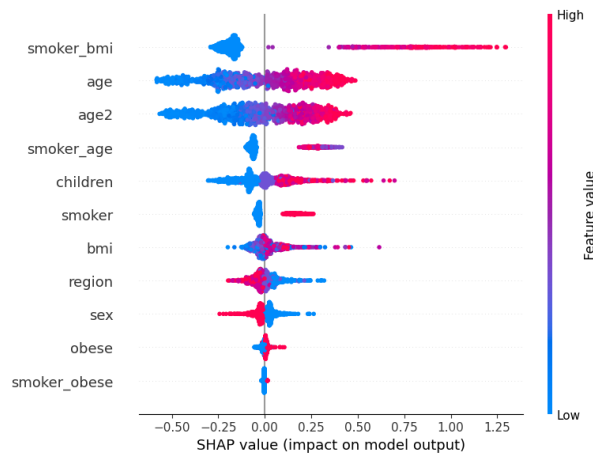


Fig. 6. SHAP value plot, which shows contributions of each datapoint within a given feature. Sorted by significance. Color represents feature value, and horizontal position represents effect on response variable (total charges).

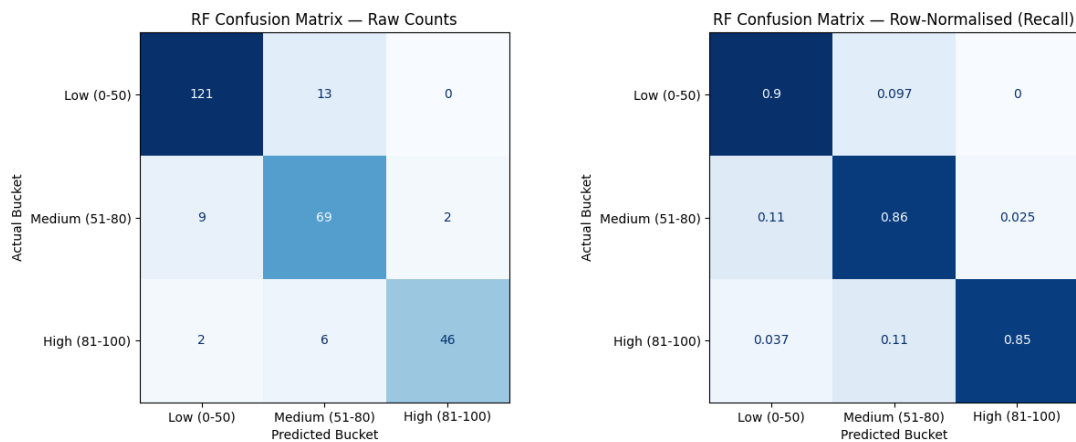


Fig. 7. Random Forest Confusion Matrix Raw and Row-Normalized

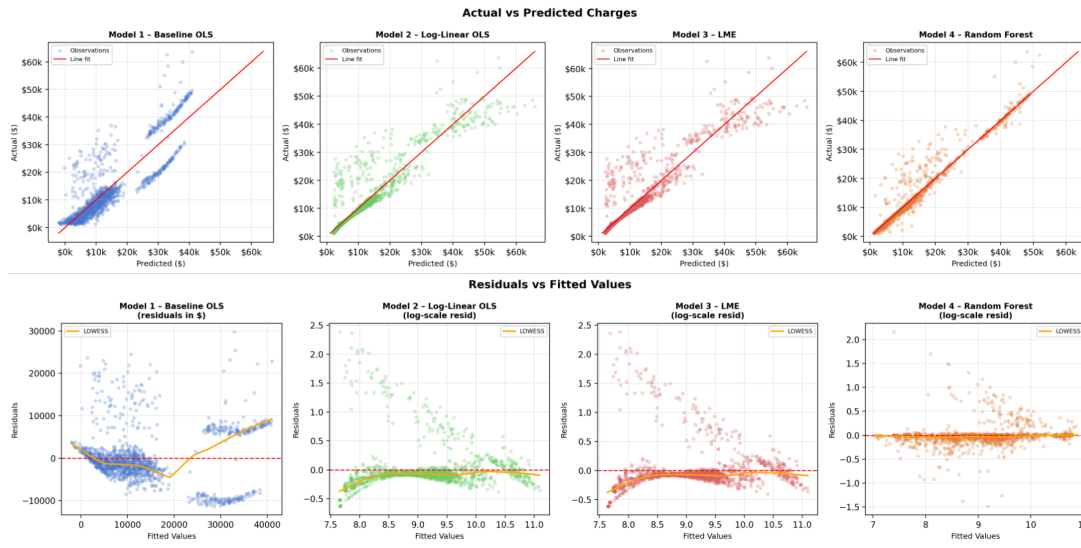


Fig. 8. Actual vs Predicted Charge & Residuals vs Fitted Values

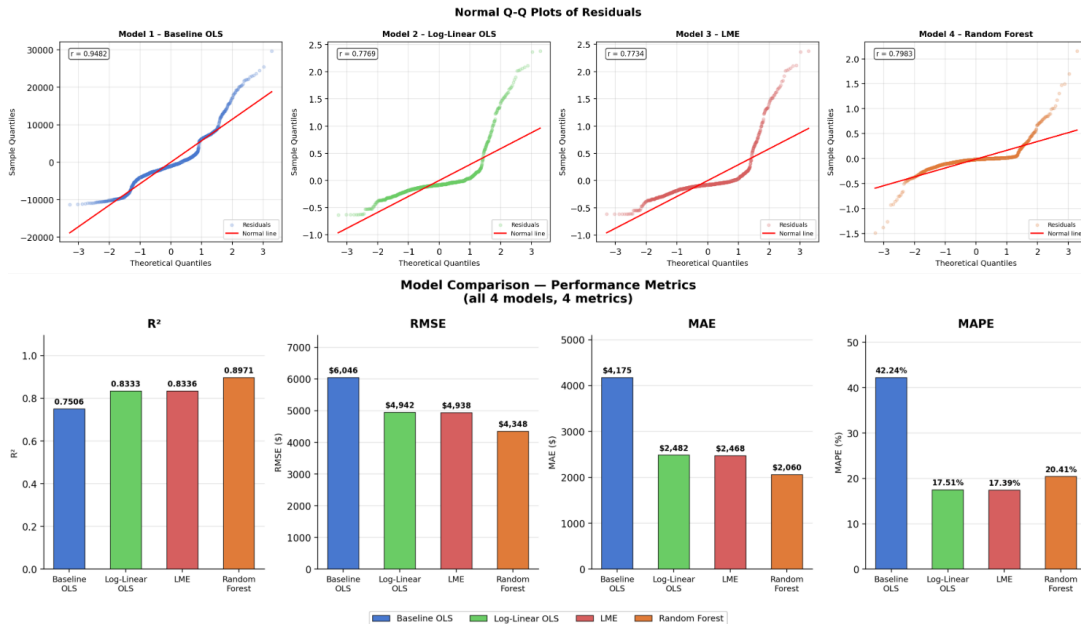


Fig. 9. Normal Q-Q Plots of Residuals & Model Comparison with Performance Metrics

B Appendix (Tables)

B.1 Baseline OLS (raw charges, base features)

OLS Regression Results						
=====						
Dep. Variable:	charges	R-squared:	0.751			
Model:	OLS	Adj. R-squared:	0.749			
Method:	Least Squares	F-statistic:	667.0			
Date:	Fri, 13 Mar 2026	Prob (F-statistic):	0.00			
Time:	14:54:07	Log-Likelihood:	-13539.			
No. Observations:	1337	AIC:	2.709e+04			
Df Residuals:	1330	BIC:	2.713e+04			
Df Model:	6					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-1.181e+04	955.719	-12.359	0.000	-1.37e+04	-9936.571
age	257.2032	11.899	21.616	0.000	233.861	280.546
sex	-129.4009	333.059	-0.389	0.698	-782.779	523.978
bmi	332.5957	27.733	11.993	0.000	278.191	387.000
children	478.7717	137.732	3.476	0.001	208.576	748.967
smoker	2.382e+04	412.051	57.806	0.000	2.3e+04	2.46e+04
region	-354.0097	151.995	-2.329	0.020	-652.185	-55.834
=====						
Omnibus:	298.466	Durbin-Watson:	2.088			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	711.712			
Skew:	1.206	Prob(JB):	2.84e-155			
Kurtosis:	5.637	Cond. No.	296.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

B.2 Log-Linear OLS (log target, age quadratic, smoker interactions)

OLS Regression Results						
=====						
Dep. Variable:	charges_log	R-squared:	0.831			
Model:	OLS	Adj. R-squared:	0.830			
Method:	Least Squares	F-statistic:	594.2			
Date:	Fri, 13 Mar 2026	Prob (F-statistic):	0.00			
Time:	14:54:07	Log-Likelihood:	-592.74			
No. Observations:	1337	AIC:	1209.			
Df Residuals:	1325	BIC:	1272.			
Df Model:	11					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	6.8323	0.121	56.687	0.000	6.596	7.069
age	0.0599	0.005	11.722	0.000	0.050	0.070
age2	-0.0002	6.36e-05	-3.632	0.000	-0.000	-0.000
sex	-0.0858	0.021	-4.120	0.000	-0.127	-0.045
bmi	0.0005	0.003	0.161	0.872	-0.006	0.007
children	0.0969	0.009	10.742	0.000	0.079	0.115
smoker	2.0081	0.190	10.578	0.000	1.636	2.381
smoker_bmi	0.0184	0.007	2.658	0.008	0.005	0.032
smoker_age	-0.0335	0.002	-18.112	0.000	-0.037	-0.030
smoker_obese	0.5134	0.087	5.927	0.000	0.343	0.683
obese	0.0026	0.039	0.066	0.947	-0.073	0.078
region	-0.0532	0.010	-5.595	0.000	-0.072	-0.035


```
=====
Omnibus:                929.736    Durbin-Watson:                1.971
Prob(Omnibus):           0.000    Jarque-Bera (JB):          10437.378
Skew:                    3.233    Prob(JB):                  0.00
Kurtosis:                15.064    Cond. No.                  4.10e+04
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
 [2] The condition number is large, 4.1e+04. This might indicate that there are strong multicollinearity or other numerical problems.

B.3 Linear Mixed-Effects (region as random intercept)

```
=====
Mixed Linear Model Regression Results
=====
Model:                MixedLM Dependent Variable: charges_log
No. Observations: 1337    Method:                REML
No. Groups:           4    Scale:                0.1434
Min. group size: 324    Log-Likelihood:        -644.6568
Max. group size: 364    Converged:              Yes
Mean group size: 334.2

-----
                Coef.  Std.Err.    z    P>|z| [0.025 0.975]
-----
const           6.741    0.126  53.367 0.000   6.494   6.989
age             0.060    0.005  11.706 0.000   0.050   0.070
age2          -0.000    0.000  -3.629 0.000  -0.000  -0.000
sex            -0.086    0.021  -4.117 0.000  -0.127  -0.045
bmi             0.001    0.003   0.307 0.759  -0.005   0.007
children        0.097    0.009  10.694 0.000   0.079   0.114
smoker          2.002    0.190  10.528 0.000   1.629   2.375
smoker_bmi      0.019    0.007   2.675 0.007   0.005   0.032
smoker_age     -0.033    0.002 -18.009 0.000  -0.037  -0.030
smoker_obese    0.510    0.087   5.874 0.000   0.340   0.680
obese           0.001    0.039   0.023 0.982  -0.075   0.077
Group Var       0.005    0.011

=====
Var(random intercept) sigma^2_u : 0.0045
Var(residual) sigma^2_e : 0.1434
ICC = sigma^2_u/(sigma^2_u+sigma^2_e) : 0.0306

RANDOM EFFECTS (region-level intercepts):
0          : u_hat = +0.0778
1          : u_hat = +0.0270
2          : u_hat = -0.0446
3          : u_hat = -0.0602
```

B.4 Random Forest Model

```
=====
Random Forest
=====

Feature importances:
smoker_bmi    0.284
age           0.208
age2          0.172
smoker_age    0.097
bmi           0.085
smoker        0.058
children      0.051
```

```
region      0.025
sex         0.012
obese       0.005
smoker_obese 0.002
```

```
R2 Train ($ scale): 0.957
R2 Test  ($ scale): 0.897
```

```
CV R2 scores: [0.763 0.827 0.751 0.78 0.781]
Mean CV R2: 0.781
Std CV R2: 0.026
```

C Appendix (Code)

Medical Cost Prediction in the United States Python Code